



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	DDA4230
Course title (English)	Reinforcement Learning
Course title (Chinese)	强化学习
Units	3
Language of Instruction	English
Description (English)	This course provides a basic introduction to reinforcement learning algorithms and their applications. Topics include multi-armed bandits; finite Markov decision processes; dynamic programming; Monte-Carlo methods; temporal-difference learning; actor-critic methods; off-policy learning; introduction to deep variants of the aforementioned algorithms, including deep Q-learning, policy gradient methods, and actor-critic methods.
Description (Chinese)	本課程是教授強化學習概念之基礎課程，透過修習本課程，可以學習強化學習上之基礎概念、理論、算法、及應用，並可以學習如何實現和使用強化學習模型。本課程主題包括多臂老虎機、有限馬爾可夫決策、動態規劃、蒙特卡洛法、時間-差分學習、PG算法、Q算法、actor-critic算法、off-policy算法、以及近似算法簡介。

2. Prerequisites / Co-requisites

A. Prerequisites

Remark:

This course would presume some basics knowledge of

MAT1001 or MAT1010 or MAT1011

MAT1002 or MAT3010 or MAT1012

MAT2040 Linear Algebra or MAT2041

CSC1001 or CSC1003

STA2001 Probability and Statistics I or STA2003 or STA2001H

STA2002 Probability and Statistics II or STA2004 or STA2002H

B. Co-requisites



3. Learning Outcomes

Knowledge:

- Students will understand the key models and concepts of reinforcement learning.
- Students will understand the strengths and limitations of popular reinforcement learning techniques.
- Students will understand popular reinforcement learning algorithms to solve the real-world problems.
- Students will be able to identify promising applications of reinforcement learning.

Skills:

- Students will utilize a programming language to learn, visualize, and mine new insights from a variety of sequential decision problems using reinforcement learning.
- Students will utilize existing software to analyze available tasks to inform critical decisions.
- Students will study decision problems scientifically and use it to derive effective methods.
- Students will be able to actively manage and participate in reinforcement learning projects executed by consultants or specialists in reinforcement learning.

Valued/Attitude:

- Students will be equipped with some reinforcement learning knowledge and be able to communicate with domain experts to solve their decision science problems.
- Students will be more vigilant in some decision science issues and aware with the decision science development in the society.
- Students will have awareness of the impact of reinforcement learning in social, industrial, environmental and technological context.
- Students will be more literate in decision science. They will develop some knowledge in decision science so that they can disseminate reinforcement science related new development.

4. Course syllabus



1. Review of probability, statistics, calculus, linear algebra, algorithms, Markov chain, and machine learning.
2. Bandit algorithms: greedy algorithms, explore then commit, upper confidence bound, Thompson sampling, lower bounds.
3. Markov decision process: formulation and definition, properties, Markov chain, Markov process and its solution and covering time, MDP examples, policy function and value function.
4. Dynamic programming: dynamic programming, policy iteration, value iteration, search algorithms.
5. Monte Carlo methods: MC policy evaluation, policy gradient, off-policy learning.
6. Temporal-difference methods: SARSA, eligibility traces, Q-learning.
7. Deep variants of the algorithms: deep Q-learning, policy gradient methods, and actor-critic methods.
8. Imitation learning methods: including dagger and inverse reinforcement learning.
9. Applications and recent advancements of reinforcement learning, including monte-carlo tree search, reinforcement learning from human feedback (preference-based reinforcement learning) and reinforcement learning in embodied AI.

5. Assessment Scheme

Component/ method	% weight
Assignments (written and coding homework)	30
Midterm exam	20
Final project	50

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B+/B/B-	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.



C+/C/C-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D+/D/D-	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials. Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

- A course questionnaire answered by every student after finishing all the lectures
- Reflections from teachers of more advanced courses
- Feedback from informal conversation over (free) coffee in my office and e-mail correspondence with students after class

8. Reading

Required Reading

1.Reinforcement learning: An introduction, 978-0262039246, Sutton, Richard S., and Andrew G. Barto, MIT press, 2ND, 2018, United States

9. Course components

Activity	Hours/week
Lecture	3
Office hours	2
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Markov decision process, Optimality of MDPs
002	Stochastic multi-armed bandits, Greedy algorithms
003	Explore-then-commit algorithms, UCB algorithms



004	Thompson sampling, Hardness of Bandits
005	Discrete MDPs, Iterative Methods
006	UCVI and PSRL, Q-Learning
007	Model-Free Policy Evaluation, Advanced Topic: Monte-Carlo Tree Search
008	Trial and Error, Value function Approximation
009	Deep Q-learning, Policy Gradient
010	Policy Optimization
011	Interconnections between policy and value
012	Imitation Learning
013	Advanced Topic: Reinforcement Learning from Human Feedback
014	Advanced Topic: Embodied AI

11. Any other information

12. Version date

Version number	4
As of (date)	2025/07/21